WORKED SOLUTION

STEP III 2005, Question 3

Pure • Composition of Quadratics

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s3-q3

The key idea

Composing two monic quadratics gives a monic quartic whose coefficients are tied together: four parameters p, q, r, s have to produce four coefficients a, b, c, d, but p and s only ever appear in the combination $2s + p$ in two of them, so one relation must hold. The second form is really the same thing in disguise: completing the square on the *outer* quadratic turns $\mathrm{f}(\mathrm{g}(x))$ into (a square) − (a constant). That form is exactly what lets us solve the final quartic with nothing more than the quadratic formula.

Solution

Composing the quadratics

Substituting $\mathrm{g}(x)$ for x in f,

$$\begin{aligned}\mathrm{f}(\mathrm{g}(x)) &= (x^2 + rx + s)^2 + p(x^2 + rx + s) + q \\ &= x^4 + 2rx^3 + (r^2 + 2s + p)x^2 + r(2s + p)x + (s^2 + ps + q).\end{aligned}$$

So $x^4 + ax^3 + bx^2 + cx + d = \mathrm{f}(\mathrm{g}(x))$ for all x exactly when

$$a = 2r, \qquad b = r^2 + (2s + p), \qquad c = r(2s + p), \qquad d = s^2 + ps + q.$$

Necessity. If such p, q, r, s exist, then $r = \frac{1}{2}a$ and $2s + p = b - r^2 = b - \frac{1}{4}a^2$, so

$$c = \tfrac{1}{2}a(b - \tfrac{1}{4}a^2), \qquad \text{that is} \qquad a^3 - 4ab + 8c = 0.$$

Sufficiency. Conversely, suppose $a^3 - 4ab + 8c = 0$. Take $r = \frac{1}{2}a$, take s to be anything (say $s = 0$), and then put $p = b - \frac{1}{4}a^2 - 2s$ and $q = d - s^2 - ps$. The first, second and fourth equations hold by construction, and the third reads $c = \frac{1}{2}a(b - \frac{1}{4}a^2)$, which is our condition rearranged. Notice that d is never constrained, since q appears only in the constant term.

So the required necessary and sufficient condition is

$$\boxed{a^3 - 4ab + 8c = 0}$$

(with d arbitrary).

The form $(x^2 + vx + w)^2 - k$

Ask yourself How can a sum like $\mathrm{g}^2 + p\,\mathrm{g} + q$ be rewritten as a single square minus a constant?

Complete the square in g, treating $\mathrm{g}(x)$ as a single variable:

$$\mathrm{f}(\mathrm{g}(x)) = (\mathrm{g}(x) + \tfrac{1}{2}p)^2 + q - \tfrac{1}{4}p^2 = (x^2 + rx + s + \tfrac{1}{2}p)^2 - (\tfrac{1}{4}p^2 - q).$$

Now we can prove both directions.

If the condition holds, the first part gives p, q, r, s with the quartic equal to $\mathrm{f}(\mathrm{g}(x))$, and the display above shows it equals $(x^2 + vx + w)^2 - k$ with

$$v = r, \qquad w = s + \tfrac{1}{2}p, \qquad k = \tfrac{1}{4}p^2 - q.$$

Conversely, $(x^2 + vx + w)^2 - k$ is itself $\mathrm{f}(\mathrm{g}(x))$ with $\mathrm{g}(x) = x^2 + vx + w$ and $\mathrm{f}(x) = x^2 - k$ (that is, $p = 0$, $q = -k$, $r = v$, $s = w$). By the necessity argument of the first part, the condition $a^3 - 4ab + 8c = 0$ must then hold.

This completes the "if and only if". For actual use it helps to have the values explicitly: expanding,

$$(x^2 + vx + w)^2 - k = x^4 + 2vx^3 + (v^2 + 2w)x^2 + 2vw\,x + (w^2 - k),$$

so we need

$$v = \tfrac{1}{2}a, \qquad w = \tfrac{1}{2}(b - \tfrac{1}{4}a^2) = \tfrac{1}{8}(4b - a^2), \qquad k = w^2 - d,$$

and then the x coefficient is $2vw = \tfrac{1}{2}a(b - \tfrac{1}{4}a^2)$, which equals c precisely when the condition holds.

Solving the quartic

Here $a = -4$, $b = 10$, $c = -12$, $d = 4$, and

$$a^3 - 4ab + 8c = -64 + 160 - 96 = 0,$$

so the method applies. The formulas give

$$v = -2, \qquad w = \tfrac{1}{8}(40 - 16) = 3, \qquad k = 3^2 - 4 = 5,$$

and indeed $(x^2 - 2x + 3)^2 - 5 = x^4 - 4x^3 + 10x^2 - 12x + 4$ (worth expanding once to be sure). The equation becomes

$$(x^2 - 2x + 3)^2 = 5 \quad \Longleftrightarrow \quad x^2 - 2x + 3 = \sqrt{5} \ \text{ or } \ x^2 - 2x + 3 = -\sqrt{5}.$$

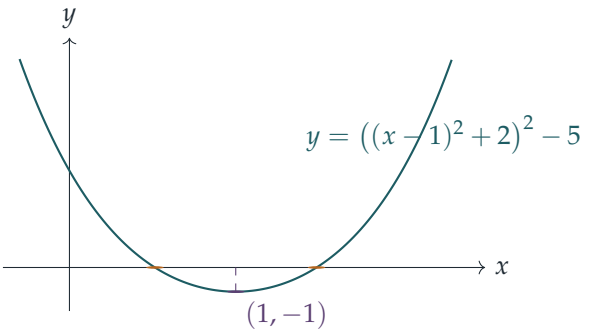
Writing $x^2 - 2x + 3 = (x - 1)^2 + 2$:

- $(x-1)^2 = \sqrt{5}-2$. Since $\sqrt{5} > 2$ the right-hand side is positive, giving two real roots $x = 1 \pm \sqrt{\sqrt{5}-2}$.
- $(x-1)^2 = -2-\sqrt{5} < 0$, giving the complex pair $x = 1 \pm \mathrm{i}\sqrt{2+\sqrt{5}}$.

So the roots are

$$\boxed{x = 1 \pm \sqrt{\sqrt{5}-2}, \qquad x = 1 \pm \mathrm{i}\sqrt{2+\sqrt{5}}}$$

with only the first pair real (if the question is read as asking for real roots, these are the two).



The graph is symmetric about $x = 1$ (it depends only on $(x-1)^2$), with minimum value $2^2 - 5 = -1$ at $x = 1$, so it crosses the axis exactly twice.

Checks

- Numerically, $1 \pm \sqrt{\sqrt{5}-2} \approx 1.4859$ and 0.5141; substituting either into $x^4 - 4x^3 + 10x^2 - 12x + 4$ gives zero to within 10^{-14}, as does $1 \pm 2.0582\,\mathrm{i}$.
- Sum of roots: the four roots add to $4 = -a$, as they must.
- Try a composition you build yourself, e.g. $\mathrm{f}(x) = x^2 + x$, $\mathrm{g}(x) = x^2 + 2x$: you get $x^4 + 4x^3 + 5x^2 + 2x$, and $a^3 - 4ab + 8c = 64 - 80 + 16 = 0$.

Common slips

- Including a condition on d: the constant term can always be matched using q (or k), so d is free.
- Proving only one direction of the "if and only if"; the observation that $(x^2 + vx + w)^2 - k$ is itself of the form $\mathrm{f}(\mathrm{g}(x))$ gives the converse in one line.
- Stopping after the real roots without noticing that the other factor gives a complex pair.

Beyond STEP

The condition $a^3 - 4ab + 8c = 0$ has a neat meaning: substituting $x = y - \frac{1}{4}a$ removes the cubic term, and the condition says exactly that the resulting *depressed* quartic has no y term, so it is a quadratic in y^2. That is the starting point of Ferrari's general solution of the quartic, and more broadly of the question of when a polynomial decomposes as a composition, which leads into Galois theory. Allenby's *Rings, Fields and Groups* is a good place to see the algebra behind which polynomial equations can be solved by radicals.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.